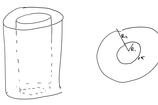
2017-01-07
Cognome: Nome:

1)



a) $h < r < R_1$ $\quad \vec{E} = 0$ (conduttore)

h $\quad R_1 < r < R_2$

TH GAUSS $\quad Q_{int} = \sigma\, 2\pi R_1 L$

$$\vec{E} = E\,\hat{u}_r$$

$$\oiint E\,\hat{u}_r\,\hat{u}_r\,dS = \frac{\sigma\, 2\pi R_1 L}{\varepsilon_0}$$

$$E\,2\pi r L = \frac{\sigma\, 2\pi R_1 L}{\varepsilon_0}$$

$$\vec{E} = \frac{\sigma R_1}{\varepsilon_0 r}\,\hat{u}_r$$

h $\quad r > R_1$

$$Q_{int} = \sigma\, 2\pi R_1 L - \sigma\, 2\pi R_2 L$$
$$= \sigma\, 2\pi L (R_1 - R_2)$$

$$E\,2\pi r L = \sigma\, 2\pi L \frac{(R_1 - R_2)}{\varepsilon_0}$$

$$\vec{E} = \frac{\sigma(R_1 - R_2)}{\varepsilon_0 r}\,\hat{u}_r$$

$$\Rightarrow \quad \vec{E} = \begin{cases} 0 & r < R_1 \\ \frac{\sigma R_1}{\varepsilon_0 r}\,\hat{u}_r & R_1 < r < R_2 \\ \frac{\sigma(R_1 - R_2)}{\varepsilon_0 r}\,\hat{u}_r & r > R_2 \end{cases}$$

b) $$V(R_2) - V(R_1) = -\int_{R_1}^{R_2} \frac{\sigma R}{\varepsilon_0 r}\,dr$$

$$= -\frac{\sigma R}{\varepsilon_0}\Big[\ln r\Big]_{R_1}^{R_2}$$
$$= -\frac{\sigma R}{\varepsilon_0}\,\ln\left(\frac{R_2}{R_1}\right)$$

$$V(R_2) - V(R_1) = \frac{\sigma R_1}{\varepsilon_0}\,\ln\left(\frac{R_1}{R_2}\right) < 0$$

$$R_1 < R_2 \Rightarrow \ln\left(\frac{R_1}{R_2}\right) < 0$$

c) $$U = \frac{1}{2} Q \cdot \Delta V = \frac{1}{2} C \Delta V^2$$
$$C = \frac{Q}{\Delta V} \qquad Q = C\Delta V$$

$$C = \frac{\sigma\, 2\pi R L}{\frac{\sigma R}{\varepsilon_0}\,\ln\left(\frac{R_2}{R_1}\right)} = \frac{2\pi\varepsilon_0 L}{\ln\left(\frac{R_2}{R_1}\right)}$$

$$U = \frac{1}{2}\,\frac{\sigma\, 2\pi R L}{\ln\left(\frac{R_2}{R_1}\right)} \cdot \frac{\sigma^2 R^2}{\varepsilon_0}\,\ln\left(\frac{R_2}{R_1}\right)$$

$$\frac{U}{L} = \frac{\pi\sigma^2 R^2}{\varepsilon_0 \cancel{\ln\left(\frac{R_2}{R_1}\right)}} \cdot \cancel{\ln\left(\frac{R_2}{R_1}\right)}$$

$$\frac{U}{L} = \frac{\pi\sigma^2 R^2}{\varepsilon_0}\,\ln\left(\frac{R_2}{R_1}\right)$$

Oppure

$$a = \frac{1}{2}\,\vec{D}\cdot\vec{E} \qquad D = \varepsilon_0 E$$

$$U = \frac{1}{2}\iiint_\tau \varepsilon E^2\,d\tau = \frac{\varepsilon_0}{2}\iiint_\tau \frac{\sigma^2 R^2}{\varepsilon_0^2 r^2}\,d\tau$$

$$= \frac{\sigma^2 R^2}{2\varepsilon_0}\left(\int_V \frac{1}{r^2}\right) \qquad d\tau = dz\,dy\,dx \quad \begin{cases} r = R \text{ cost} \\ \theta = \pi \text{ cost} \\ z = t \end{cases} \quad d\tau = r\,dz\,d\theta\,dr$$

$$\frac{\sigma^2 R^2}{2\varepsilon_0}\left(2\pi \int_0^L \left(\int_{R_1}^{R_2} \frac{1}{r}\,dr\right) dz\right) \qquad \Rightarrow \frac{\sigma^2 R^2}{\varepsilon_0}\,L\,\ln\frac{R_2}{R_1} \qquad \Rightarrow \quad U$$

$$\frac{U}{L} = \frac{\sigma^2 R^2}{\varepsilon_0}\,\ln\frac{R_2}{R_1} \qquad \text{ok} \qquad \text{ok}$$

2) a) Leggi di Maxwell

$$\oint_S \vec{B}\cdot d\vec{l} = \mu_0 \sum_i I_{c_i}$$

Dove $\sum_i I_{c_i}$ è intesa tutte le correnti concentrate al circuito σ

I flusso $\vec{B}$ è

$$\oint_{\sigma = \partial S} \vec{B}\cdot d\vec{l} = \iint_S \vec{\nabla}\times\vec{B}\cdot\hat{n}\,dS$$

$$\iint_S (\vec{\nabla}\times\vec{B})\cdot\hat{n}\,dS = \mu_0 \sum I_c$$

$$\sum I_c = \iint_S \vec{J}_c\cdot n\,dS$$

$$\Rightarrow \iint_S (\nabla\times B)\cdot\hat{n}\,dS = \iint_S \mu_0 \vec{J}_c\cdot\hat{n}\,dS$$

$$\Leftrightarrow \quad \vec{\nabla}\times\vec{B} = \mu_0\,\vec{J}_c$$

$\boxed{\text{I}}$ eq Maxwell stazionaria

Tale equazione corrisponde con la $\boxed{\text{I}}$ eq Maxwell

Il Biot implicava che il campo di induzione magnetica sia sempre chiuso e e si avvalga attorno alle correnti

$\boxed{\text{II}}$ eq Maxwell non stazionaria

$$\vec{\nabla}\times\vec{B} = \mu_0\left(\vec{J}_c + \varepsilon_0\frac{\partial\vec{E}}{\partial t}\right)$$

Dove da vedere

3) a) $\int B_n] = 0 \qquad [B_n] = 0$

$[E_t] = 0 \qquad [H_t] = J_s$ cond

b) $\int$ diffusione

Riflettenza $\quad R = \frac{E_r^2}{E_i^2}$

Trasmittenza $\quad T = \frac{n_2}{n_1}\frac{E_t^2}{E_i^2}$

Per il conservazione dell'energia $\quad R + T = 1$

$\int$ l'aria è il dielettrico

$\int$ a da $n_1 = 1 \qquad n_2 = 1.4$

$$R = \left|\frac{n_1 - n_2}{n_1 + n_2}\right|^2$$
$$= \left|\frac{1 - 1.4}{1 + 1.4}\right|^2 = \left|\frac{\sqrt{0.4}}{\sqrt{2.4}}\frac{e^{-i\pi}}{e^{-i\pi}}\right|^2 = \left|e^{i\pi}\cdot\frac{1}{6}\right|^2 = 1$$

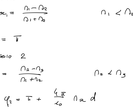
$R = 1 \Rightarrow T = 0$

$\int$ l'onda è totalmente riflessa

4) $n_1 = 1.6$

$n_2 = 1.25$

$d = 600\,nm$ vetro



a) Interferenza distruttiva in riflessione

$$\Delta\varphi = \frac{4\pi}{\lambda}\Delta L_{ott} \qquad L_{ott} = \sum_i n_i L_i$$

$$\Delta\varphi = (2m+1)\pi$$

PASSO 1

$$r_1 = \frac{n_1 - n_2}{n_1 + n_2} \qquad n_1 < n_2$$

$$\varphi_1 = \pi$$

PASSO 2

$$r_2 = \frac{n_2 - n_3}{n_2 + n_3} \qquad n_2 < n_3$$

$$\Rightarrow \varphi_2 = \pi + \frac{4\pi}{\lambda_0}\,n_2\,d$$

$$\Delta\varphi = \varphi_2 - \varphi_1 = \frac{4\pi}{\lambda_0}\,n_2\,d = (2m+1)\cancel{\pi}$$

$$d = \frac{\lambda_0}{4n_2}\,(2m+1)$$
$$= 115\,(2m+1)\cdot 10^{-9}\,m$$
$$115\,(2m+1)\,nm$$

b) $I_r = R\,I_i$

$$R = \left(\frac{n_1 n_3 - n_2^2}{n_1 n_3 + n_2^2}\right)^2 = 0 \Rightarrow \quad n_2^2 = n_1 n_3$$

$$n_2 = \sqrt{n_1 n_3} = 1.26 \quad \text{Vero}$$

NB

$$R = \left(\frac{n_2 n_1 - n_2^2}{n_2 n_1 + n_2}\right)^2$$